CV

Kanghyeok Im

Researcher, Department of Climate, Energy & Environment, Gyeonggi Research Institute

khim@gri.re.kr

I studied urban and regional planning, and I analyze urban issues with GIS and spatial data. After working on population migration, transportation, industry, and water management, I now study cities in the era of carbon neutrality and the energy transition. As a team leader and project manager on commissioned and joint research, I have taken projects from planning through to the final report.

4.27 / 4.5 M.S. GPA	**1st** Ranked in B.S. department	**450M** Administrative records processed	**9** Research works

Experience

2026.8 – Present

Department of Climate, Energy & Environment, Gyeonggi Research Institute

Researcher

- Urban research on carbon neutrality and the energy transition
- Reviewing conditions for renewable energy deployment through GIS-based spatial analysis

2022.3 – 2022.10

Small Manufacturers Support Team, Yongin Industry Promotion Agency

Intern

- Identified new industrial clusters through ICT industry statistics and QGIS visualization, helping expand the supported industries and areas
- Helped develop 3 new support programs and supported over 100 prototype productions
- Prepared program guides and public notices, and operated facilities in the urban small manufacturers cluster

QGIS · Industry statistics

Research Experience

2024.9 – 2025.1

Establishing a Korean Water Culture

Seoul National University R&DB Foundation · Client: K-water Operation & Management Co.

Project Manager · Research Assistant

- Managed all research administration, including contracts, budget, and project completion, in line with the client's internal rules
- Reviewed cases from 5 countries and Korean institutions to map stakeholders and governance structures
- Designed and analyzed a survey of 400 citizens, and wrote the report
- Defined water culture along four pillars (history & tradition, society & economy, environment & ecology, arts & leisure) and set directions for cultural facilities

Case study · Survey · Research administration

2023.12 – 2024.8

Revision of the Guidelines for Comprehensive Water Demand Management Plans

Seoul National University R&DB Foundation · Client: Korea Water Resources Corporation (K-water)

Research Assistant

- Reviewed the current state and problems of Korea's water demand management planning and evaluation system
- Reviewed systems in the US, Australia, the UK, and Singapore to derive implications for the Korean guidelines
- Drafted revised guidelines that make target setting, savings estimates, and performance evaluation more consistent
- Proposed an improved evaluation system for provinces and metropolitan cities, and wrote the proposal and reports

Policy comparison · Policy evaluation

Education

2023.9 – 2025.8

Graduate School of Environmental Studies, Seoul National University

Department of Environmental Planning, Urban & Regional Planning major · M.S. in City Planning

- GPA 4.27 / 4.5
- Focused on urban big data research, analyzing population, transportation, and industry data with GIS and econometric models
- Thesis: "Employment and Income Characteristics of Out-Migrants from Depopulating Areas"

2020.3 – 2023.2

Chung-Ang University

School of Civil and Environmental Engineering, Urban Systems Engineering major · B.S. in Engineering

- GPA 4.21 / 4.5 · Ranked 1st in Department
- Transferred from electronic engineering to study urban planning, transportation, and urban design
- Excellence Award for the capstone design project "Installation Standards for Public Truck Parking Lots Based on Utilization Rates"

Research Projects

2024.9 – 2024.12

Effects of Firm Size and Regional Characteristics on Innovation Output

Team project, Urban and Regional Analysis course

Team Leader

- Analyzed drivers of innovation output using patent applications and knowledge-based industry data by city and county
- Mapped how the link between service-firm size and innovation varies across regions

OLS · Geographically Weighted Regression (GWR) · Python

2024.4 – 2024.8

Factors Behind Changes in Bus Boardings and Alightings after a New Public Transit Policy

Joint research, BK Interdisciplinary Seminar

Team Leader

- Analyzed changes in bus boardings by Seoul neighborhood before and after the Climate Card and K-Pass
- Found floor areas of apartments, offices, neighborhood facilities, and retail significantly related to ridership change

T-test · Moran's I · Spatial Error Model (SEM)

2024.3 – 2024.6

Contributing to Carbon Neutrality through the Mobility Transition

Research report, Smart Cities and Social Innovation course

- Compared cases including Zurich, Copenhagen, Tallinn's free public transit, Germany's 9-euro ticket, and Seoul's Climate Card and K-Pass
- Proposed a transition combining transit access, walking, cycling, and car demand management

Policy case comparison

2023.9 – 2023.12

Sustainable Regional Innovation through Reuse of Vacant Houses

Course research report

- Compared Korean and international vacant-house cases by operator, revenue model, and resident participation

ESG framework · Case study

Co-authored

Green Infrastructure as a Catalyst for Urban Regeneration

Cheonggyecheon Stream restoration and private development (4 co-authors)

Data collection and preprocessing, report writing

- Tracked how much, and for how long, the stream restoration drew private development to nearby parcels

Logistic DID | K-means clustering

Awards

2025.4

Best Paper Award

Korean Planning Association Spring Conference · National and Regional Planning II session

- "Employment and Income Characteristics of Out-Migrants from Depopulating Areas"

2021.6

Capstone Design Excellence Award

Urban Systems Engineering, Chung-Ang University · Urban Planning session

- "Installation Standards for Public Truck Parking Lots Based on Utilization Rates: The Case of Cheoin-gu, Yongin-si"

Presentations

2025.4

Employment and Income Characteristics of Out-Migrants from Depopulating Areas

Korean Planning Association Spring Conference · National and Regional Planning II session

Best Paper Award

2024.8

Factors Behind Changes in Bus Boardings and Alightings after a New Public Transit Policy

BK Interdisciplinary Seminar final presentation · Graduate School of Environmental Studies, Seoul National University

Skills

Tools

Python | R | Stata | SPSS | QGIS | ArcGIS Pro | Excel

Quantitative & Spatial Analysis

Logistic Regression | Multiple Regression | CEM Matching | Staggered Difference-in-Differences (DID) | Geographically Weighted Regression (GWR) | Spatial Error Model (SEM) | Spatial Autocorrelation (Moran's I) | K-means Clustering

Regional Economic Analysis

Input-Output Analysis | Social Accounting Matrix (SAM) Model | Shift-Share Analysis | Rank Mobility Index

Data

Large-scale administrative panel construction (~450 million records) | API & Web Scraping | Survey Design & Analysis

Selected Coursework

Fall 2024	**Urban and Regional Analysis** Rank mobility index, shift-share analysis, spatial autocorrelation, and GWR using Python and GIS
Fall 2024	**Land and Housing Policy** Presented papers on the effects of real estate policy on migration and regional economies
Spring 2024	**Research Methods and Statistics** Surveys, data preprocessing and visualization, regression models, and statistical tests in R
Fall 2023	**Urban Big Data Analytics** Python, API & web scraping, K-means, treatment–control matching, and difference-in-differences
Fall 2023	**Urban and Regional Economics** Regional economic ripple effects of industrial change using input-output analysis and SAM models

Certifications & Languages

2026.7	**AIBT (AI Business Literacy Test)** Korea Productivity Center
2026.7	**AI-POT Level 1 (AI Prompt Proficiency Test)** Korea Productivity Center
2026.1	**TOEIC Speaking AL (160)** YBM
2023.3	**ADsP (Advanced Data Analytics Semi-Professional)** Korea Data Agency
2021.12	**Engineer Certificate in City Planning (Korea)** HRD Korea
2021.10	**Computer Specialist in Spreadsheet & Database, Level 1** Korea Chamber of Commerce & Industry

Training

2025.1	**ESG Education Academy** ESG Youth Forum · 24 hours
2022.2	**Data Science with R** Statistics Research Institute, Seoul National University · 20 hours
2020.8	**Online Practical Training in Traffic Accident Analysis** Korea Road Traffic Authority · 8 hours

Activities

2021.7 – 2023.12

Yongin Youth Network

Head of Jobs Subcommittee · Member of Urban Regeneration Subcommittee

- Helped shift a youth suit-rental program from businesses outside the city to businesses within Yongin
- Helped prepare a software support program for young people
- Took part in the Jungang-dong urban regeneration plan and proposed additional public transit routes